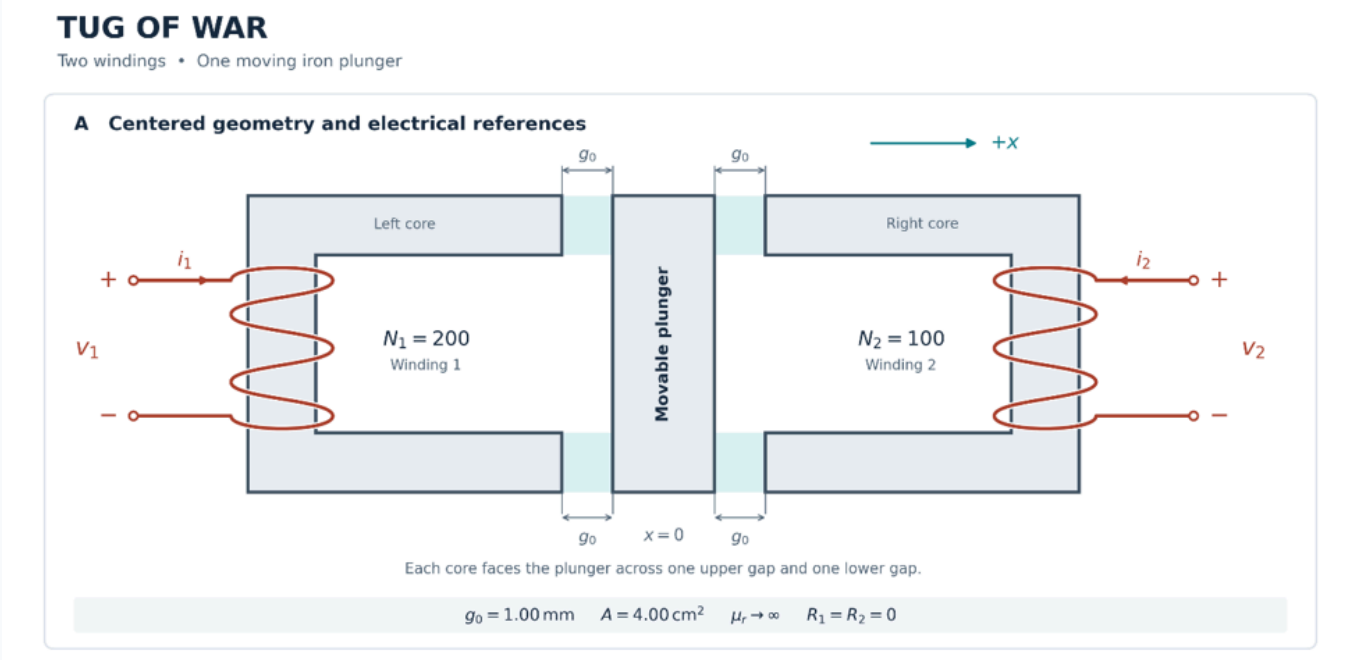
# 1. Define the variables and reference directions




**System description**

Two fixed C-shaped iron cores face a central movable iron plunger. Winding 1 is on the left core, and winding 2 is on the right core. Each core faces the plunger across two air gaps: one upper gap and one lower gap.

For horizontal motion, define the plunger displacement x as positive to the right. At x = 0, all four air gaps have length $g_0$.

The parameters are:

$N_1 = 200$ turns
$N_2 = 100$ turns
$g_0 = 1.00$ mm
$A = 4.00$ cm²
$\mu_0 = 4\pi \times 10^{-7}$ H/m

Use the following assumptions:

- All iron has infinite permeability.
- Both winding resistances are zero.
- Leakage, hysteresis, and eddy currents are negligible.
- Neglect fringing in the reluctance calculations.
- For horizontal motion, each gap has constant face area A.
- For vertical motion in Part 9, use the specified model for changing pole overlap.
- Neglect gravity, friction, and external mechanical forces unless a constraint is specified.
- Ideal electrical sources can supply or absorb energy.

Use the passive sign convention for each winding: positive current enters its positive voltage terminal.

Treat Parts 4, 6, 7, 9, 10, and 11 as separate experiments. Use the stated initial conditions for each experiment.

Before writing governing equations, annotate the diagram and define all relevant variables.

(a) Define the mechanical variables, including plunger displacement, velocity, and net electromagnetic force. Mark their positive directions.

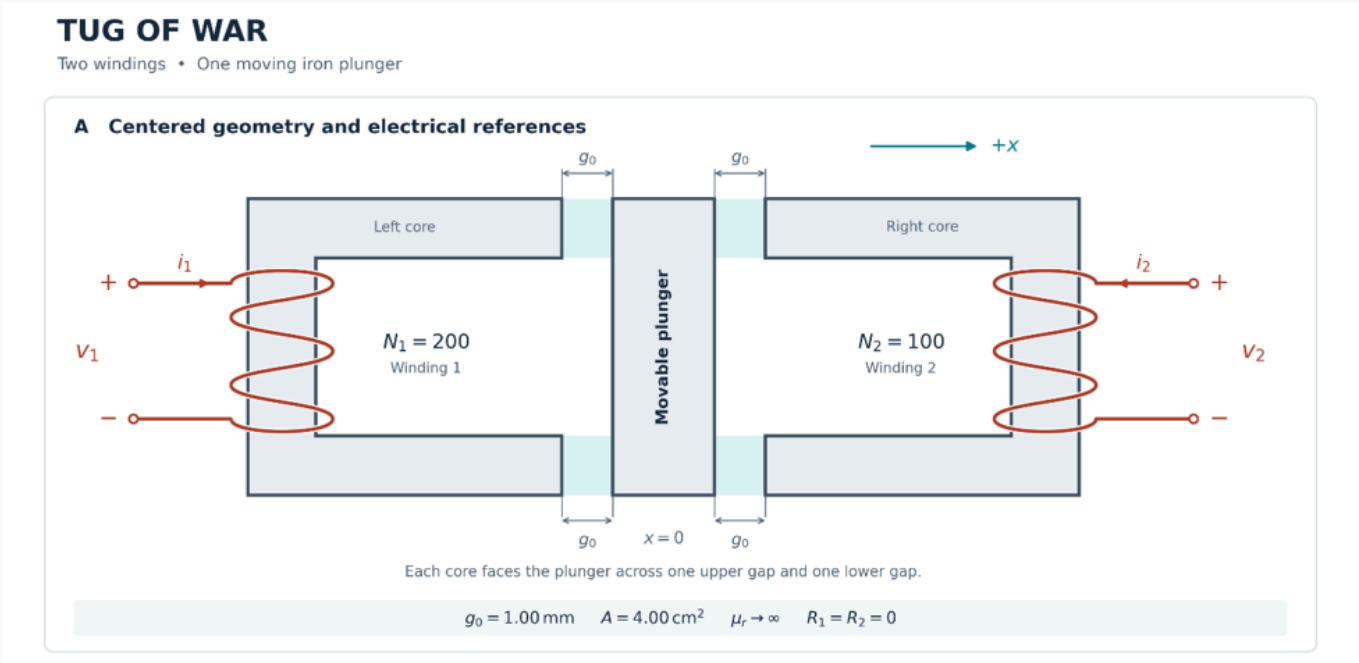
(b) Define the electrical variables, including winding currents, terminal voltages, and turn counts. Mark the current directions and voltage polarities.

(c) Define the magnetic variables, including loop fluxes, winding flux linkages, air-gap lengths, and gap-face areas. Mark consistent positive flux directions.

(d) Express all four air-gap lengths in terms of $g_0$ and x.

(e) Identify the prescribed quantities and the quantities that must be determined.

# 2. Draw the magnetic equivalent circuit




**System description**

Two fixed C-shaped iron cores face a central movable iron plunger. Winding 1 is on the left core, and winding 2 is on the right core. Each core faces the plunger across two air gaps: one upper gap and one lower gap.

For horizontal motion, define the plunger displacement x as positive to the right. At x = 0, all four air gaps have length $g_0$.

The parameters are:

$N_1 = 200$ turns
$N_2 = 100$ turns
$g_0 = 1.00$ mm
$A = 4.00$ cm²
$\mu_0 = 4\pi \times 10^{-7}$ H/m

Use the following assumptions:

- All iron has infinite permeability.
- Both winding resistances are zero.
- Leakage, hysteresis, and eddy currents are negligible.
- Neglect fringing in the reluctance calculations.
- For horizontal motion, each gap has constant face area A.
- For vertical motion in Part 9, use the specified model for changing pole overlap.
- Neglect gravity, friction, and external mechanical forces unless a constraint is specified.
- Ideal electrical sources can supply or absorb energy.

Use the passive sign convention for each winding: positive current enters its positive voltage terminal.

Treat Parts 4, 6, 7, 9, 10, and 11 as separate experiments. Use the stated initial conditions for each experiment.

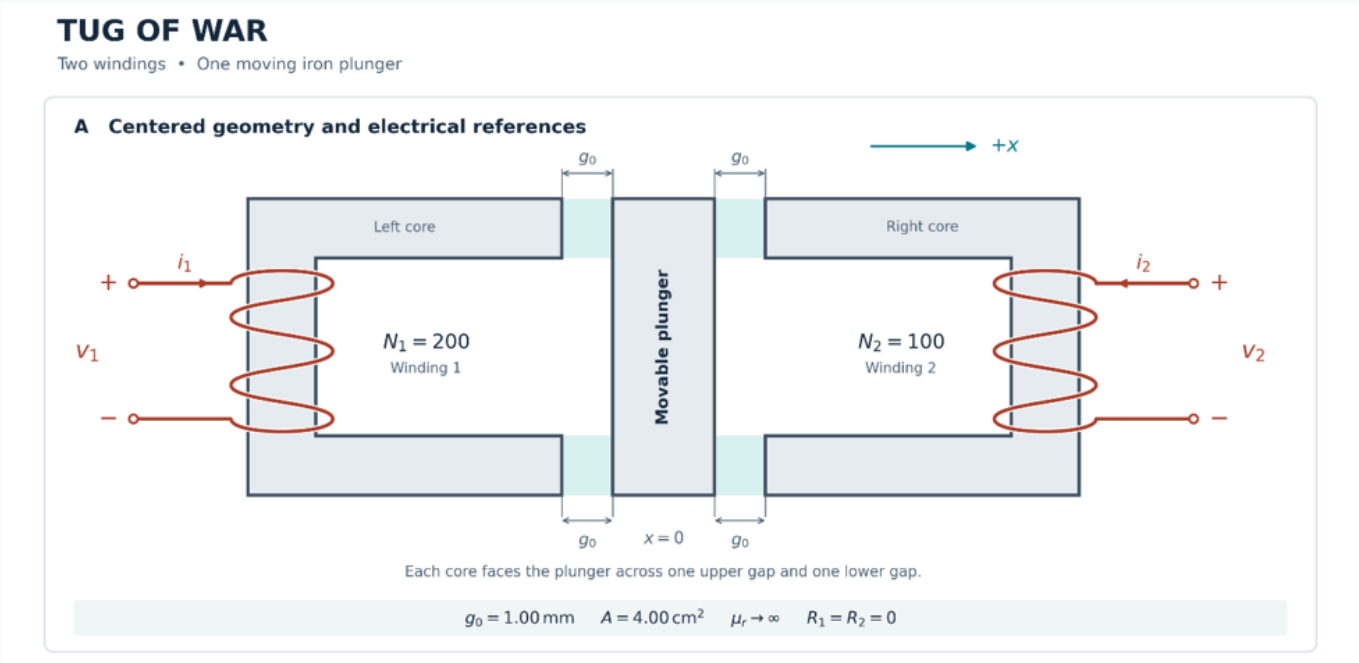
Using your reference directions from Part 1, draw the magnetic equivalent circuit.

Clearly show:

- All four air-gap reluctances, expressed in terms of x.
- Both winding MMF sources, including their polarities.
- The magnetic fluxes and their positive directions.
- The shared center iron plunger, represented consistently with the assumption of infinite iron permeability.

Write the magnetic-circuit equations needed to determine the fluxes.

# 3. Develop the flux-linkage model




**System description**

Two fixed C-shaped iron cores face a central movable iron plunger. Winding 1 is on the left core, and winding 2 is on the right core. Each core faces the plunger across two air gaps: one upper gap and one lower gap.

For horizontal motion, define the plunger displacement x as positive to the right. At x = 0, all four air gaps have length $g_0$.

The parameters are:

$N_1 = 200$ turns
$N_2 = 100$ turns
$g_0 = 1.00$ mm
$A = 4.00$ cm²
$\mu_0 = 4\pi \times 10^{-7}$ H/m

Use the following assumptions:

- All iron has infinite permeability.
- Both winding resistances are zero.
- Leakage, hysteresis, and eddy currents are negligible.
- Neglect fringing in the reluctance calculations.
- For horizontal motion, each gap has constant face area A.
- For vertical motion in Part 9, use the specified model for changing pole overlap.
- Neglect gravity, friction, and external mechanical forces unless a constraint is specified.
- Ideal electrical sources can supply or absorb energy.

Use the passive sign convention for each winding: positive current enters its positive voltage terminal.

Treat Parts 4, 6, 7, 9, 10, and 11 as separate experiments. Use the stated initial conditions for each experiment.

Using your magnetic equivalent circuit, derive the winding flux linkages:

$\lambda_1(i_1, i_2, x)$ and $\lambda_2(i_1, i_2, x)$.

(a) Express the results in terms of the self and mutual inductances:

$\lambda_1 = L_{11}(x)i_1 + L_{12}(x)i_2$

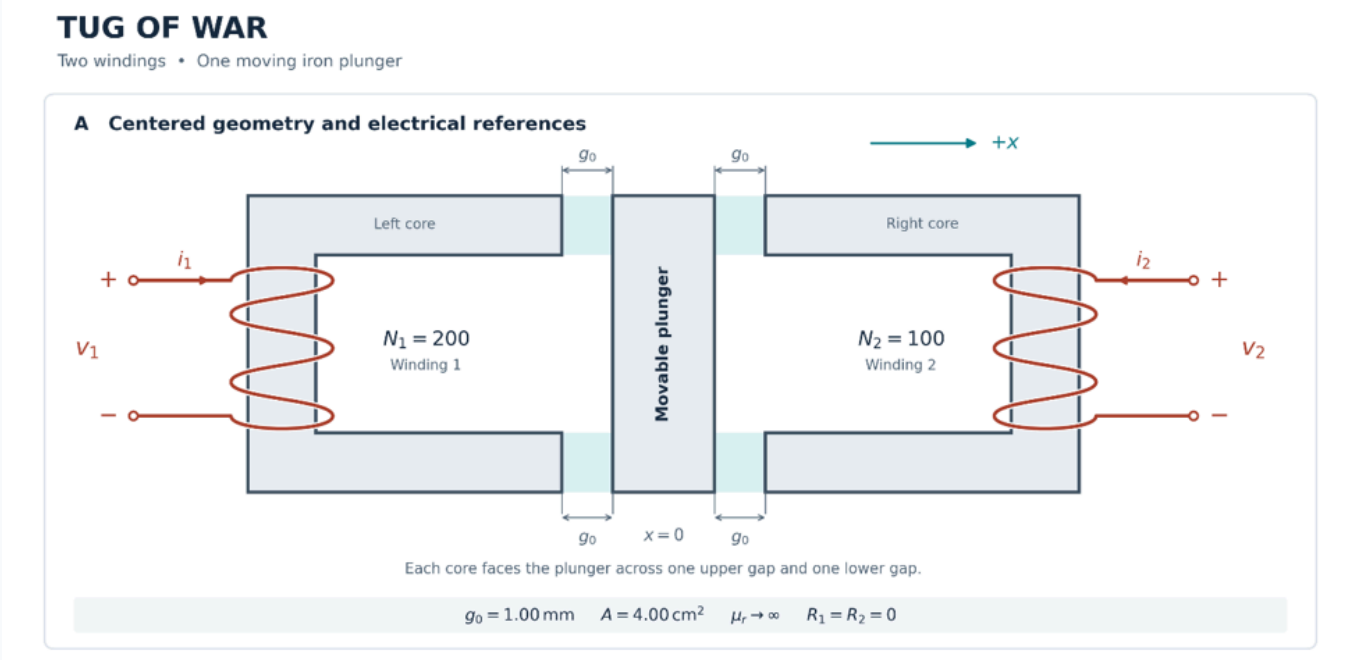
$\lambda_2 = L_{21}(x)i_1 + L_{22}(x)i_2$

(b) Write the complete inductance matrix.

(c) Identify how each inductance depends on plunger position.

(d) Does sharing the center iron plunger necessarily produce nonzero mutual inductance under the stated assumptions? Explain using your magnetic equivalent circuit.

# 4. Calculate the air-gap energies and predict the initial motion




**System description**

Two fixed C-shaped iron cores face a central movable iron plunger. Winding 1 is on the left core, and winding 2 is on the right core. Each core faces the plunger across two air gaps: one upper gap and one lower gap.

For horizontal motion, define the plunger displacement x as positive to the right. At x = 0, all four air gaps have length $g_0$.

The parameters are:

$N_1 = 200$ turns
$N_2 = 100$ turns
$g_0 = 1.00$ mm
$A = 4.00$ cm²
$\mu_0 = 4\pi \times 10^{-7}$ H/m

Use the following assumptions:

- All iron has infinite permeability.
- Both winding resistances are zero.
- Leakage, hysteresis, and eddy currents are negligible.
- Neglect fringing in the reluctance calculations.
- For horizontal motion, each gap has constant face area A.
- For vertical motion in Part 9, use the specified model for changing pole overlap.
- Neglect gravity, friction, and external mechanical forces unless a constraint is specified.
- Ideal electrical sources can supply or absorb energy.

Use the passive sign convention for each winding: positive current enters its positive voltage terminal.

Treat Parts 4, 6, 7, 9, 10, and 11 as separate experiments. Use the stated initial conditions for each experiment.

The plunger is initially centered and at rest:

$x = 0$ and $\frac{dx}{dt} = 0$.

Ideal current sources maintain:

$i_1 = 1.00$ A and $i_2 = 1.90$ A.

(a) Calculate the magnetic flux density in each air gap.

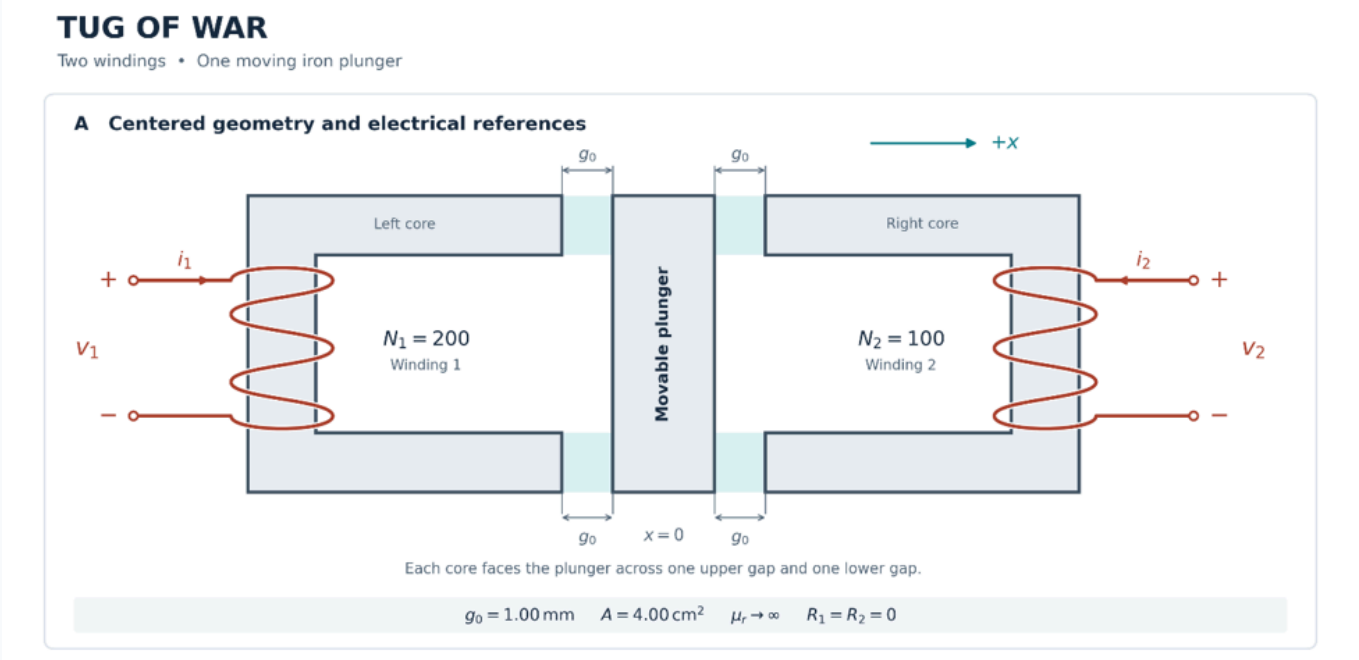
(b) Calculate the magnetic energy stored in each of the four air gaps.

(c) The plunger is released. Predict its initial direction of motion and justify your answer physically.

(d) A student claims: "Winding 2 carries more current, so it must pull harder." Evaluate this reasoning.

(e) Are stored-energy values at one position, by themselves, sufficient to determine force direction in general? Explain what additional information an energy argument requires.

# 5. Draw the equivalent electrical circuits




**System description**

Two fixed C-shaped iron cores face a central movable iron plunger. Winding 1 is on the left core, and winding 2 is on the right core. Each core faces the plunger across two air gaps: one upper gap and one lower gap.

For horizontal motion, define the plunger displacement x as positive to the right. At x = 0, all four air gaps have length $g_0$.

The parameters are:

$N_1 = 200$ turns
$N_2 = 100$ turns
$g_0 = 1.00$ mm
$A = 4.00$ cm²
$\mu_0 = 4\pi \times 10^{-7}$ H/m

Use the following assumptions:

- All iron has infinite permeability.
- Both winding resistances are zero.
- Leakage, hysteresis, and eddy currents are negligible.
- Neglect fringing in the reluctance calculations.
- For horizontal motion, each gap has constant face area A.
- For vertical motion in Part 9, use the specified model for changing pole overlap.
- Neglect gravity, friction, and external mechanical forces unless a constraint is specified.
- Ideal electrical sources can supply or absorb energy.

Use the passive sign convention for each winding: positive current enters its positive voltage terminal.

Treat Parts 4, 6, 7, 9, 10, and 11 as separate experiments. Use the stated initial conditions for each experiment.

> **Referenced Part 4 conditions**
>
> The plunger is initially centered and at rest: x = 0 and $\frac{dx}{dt} = 0$. Ideal current sources maintain $i_1 = 1.00$ A and $i_2 = 1.90$ A.

Draw separate equivalent electrical circuits for the left and right windings. Show the voltage contributions associated with changing winding current and with plunger motion.

Remember that both winding resistances are zero.

(a) Mark the current direction, terminal-voltage polarity, and voltage polarity across every element.

(b) Write the corresponding KVL equation for each winding.

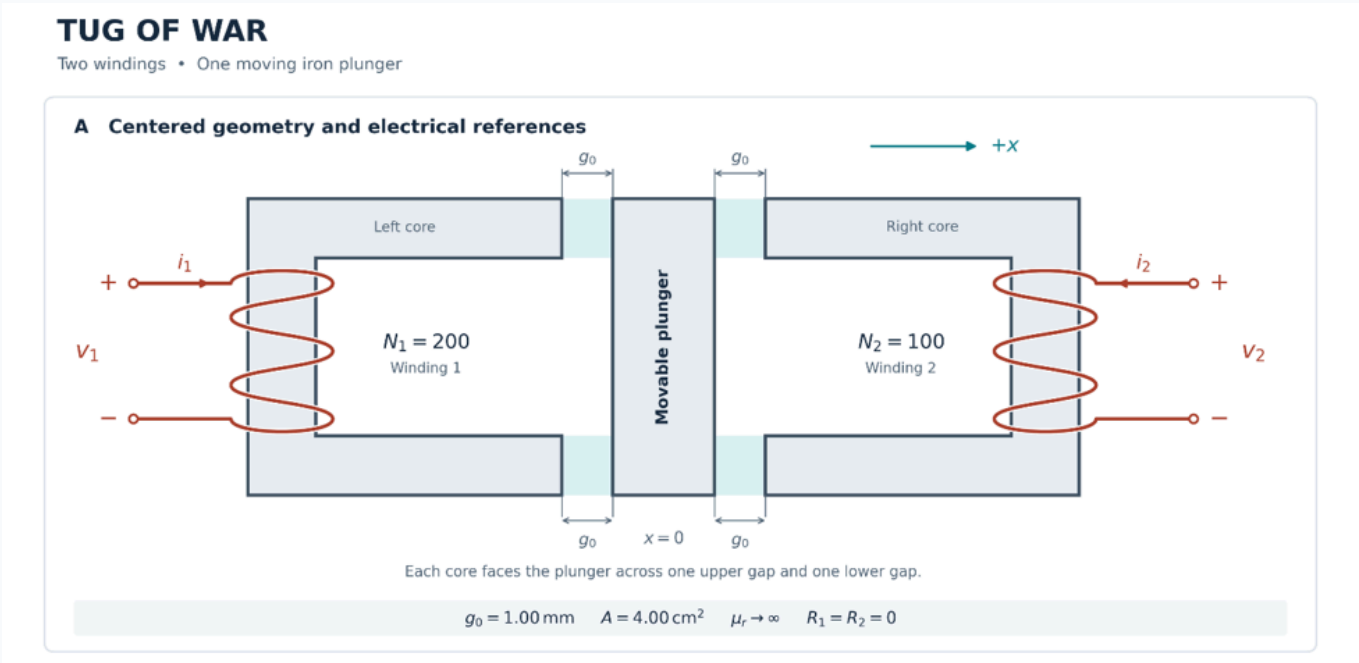
(c) Derive expressions for the current-change voltage and the speed voltage in each winding.

(d) Rearrange the equations to obtain expressions for $\frac{di_1}{dt}$ and $\frac{di_2}{dt}$ in terms of the terminal voltages, currents, position, and velocity.

(e) With both currents positive and maintained constant, determine the actual speed-voltage polarities as the plunger moves in the direction predicted in Part 4.

(f) Evaluate this statement: "The windings are stationary, so neither winding can have a speed voltage."

# 6. Predict voltage polarities with only winding 1 excited




**System description**

Two fixed C-shaped iron cores face a central movable iron plunger. Winding 1 is on the left core, and winding 2 is on the right core. Each core faces the plunger across two air gaps: one upper gap and one lower gap.

For horizontal motion, define the plunger displacement x as positive to the right. At x = 0, all four air gaps have length $g_0$.

The parameters are:

$N_1 = 200$ turns
$N_2 = 100$ turns
$g_0 = 1.00$ mm
$A = 4.00$ cm²
$\mu_0 = 4\pi \times 10^{-7}$ H/m

Use the following assumptions:

- All iron has infinite permeability.
- Both winding resistances are zero.
- Leakage, hysteresis, and eddy currents are negligible.
- Neglect fringing in the reluctance calculations.
- For horizontal motion, each gap has constant face area A.
- For vertical motion in Part 9, use the specified model for changing pole overlap.
- Neglect gravity, friction, and external mechanical forces unless a constraint is specified.
- Ideal electrical sources can supply or absorb energy.

Use the passive sign convention for each winding: positive current enters its positive voltage terminal.

Treat Parts 4, 6, 7, 9, 10, and 11 as separate experiments. Use the stated initial conditions for each experiment.

Winding 1 is supplied by an ideal constant-current source:

$i_1 = 1.00$ A.

Winding 2 is open-circuited:

$i_2 = 0$.

At the instant under consideration, the plunger is moving left.

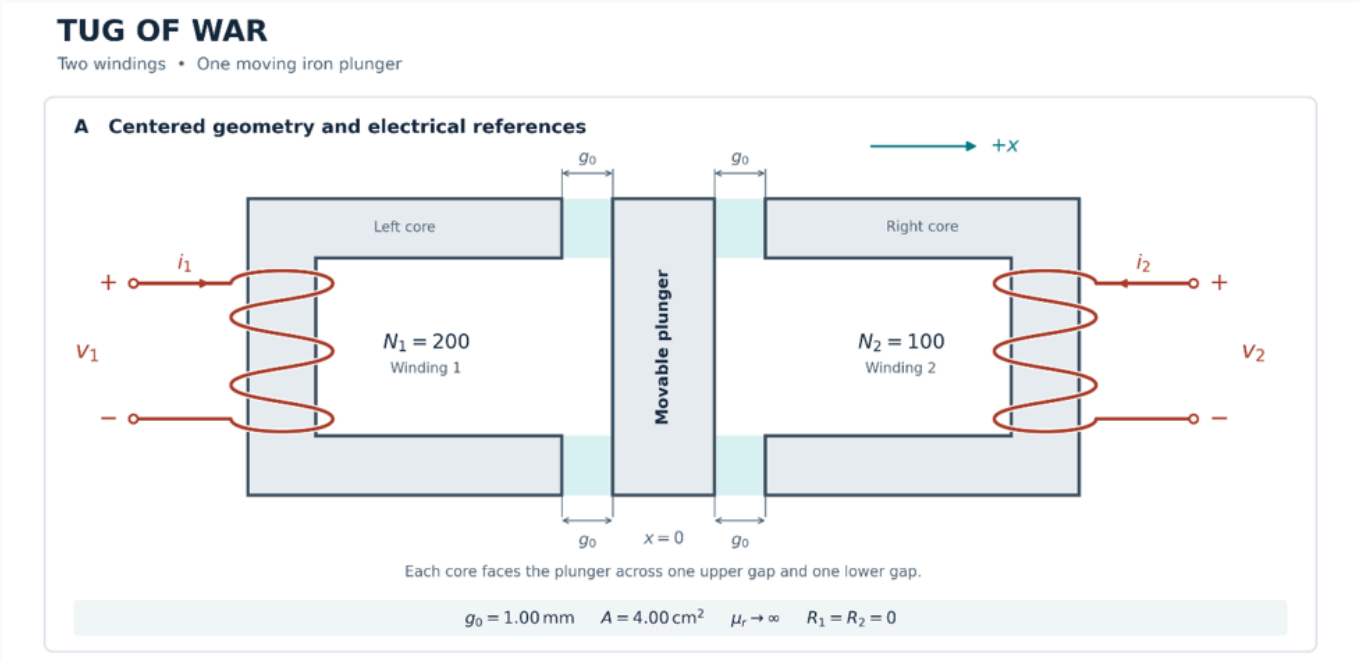
(a) What terminal-voltage polarity appears across winding 1? Identify which terminal is at the higher potential relative to your current reference.

(b) What voltage appears across winding 2?

(c) Justify both answers using the changing flux linkages.

(d) Does the fact that both windings share a moving iron plunger guarantee an induced voltage in the open winding? Explain.

# 7. Compare constant-current and constant-voltage excitation




**System description**

Two fixed C-shaped iron cores face a central movable iron plunger. Winding 1 is on the left core, and winding 2 is on the right core. Each core faces the plunger across two air gaps: one upper gap and one lower gap.

For horizontal motion, define the plunger displacement x as positive to the right. At x = 0, all four air gaps have length $g_0$.

The parameters are:

$N_1 = 200$ turns
$N_2 = 100$ turns
$g_0 = 1.00$ mm
$A = 4.00$ cm²
$\mu_0 = 4\pi \times 10^{-7}$ H/m

Use the following assumptions:

- All iron has infinite permeability.
- Both winding resistances are zero.
- Leakage, hysteresis, and eddy currents are negligible.
- Neglect fringing in the reluctance calculations.
- For horizontal motion, each gap has constant face area A.
- For vertical motion in Part 9, use the specified model for changing pole overlap.
- Neglect gravity, friction, and external mechanical forces unless a constraint is specified.
- Ideal electrical sources can supply or absorb energy.

Use the passive sign convention for each winding: positive current enters its positive voltage terminal.

Treat Parts 4, 6, 7, 9, 10, and 11 as separate experiments. Use the stated initial conditions for each experiment.

Reset the plunger to the centered position, initially at rest.

Winding 1 has an established current maintained at:

$i_1 = 1.00$ A.

Winding 2 initially carries zero current. At t = 0, it is connected to an ideal constant positive voltage $V_2$, measured using its passive voltage reference.

(a) In which direction does the plunger initially move? Explain what happens to the current in winding 2 immediately after connection.

(b) Derive $\lambda_2(t)$.

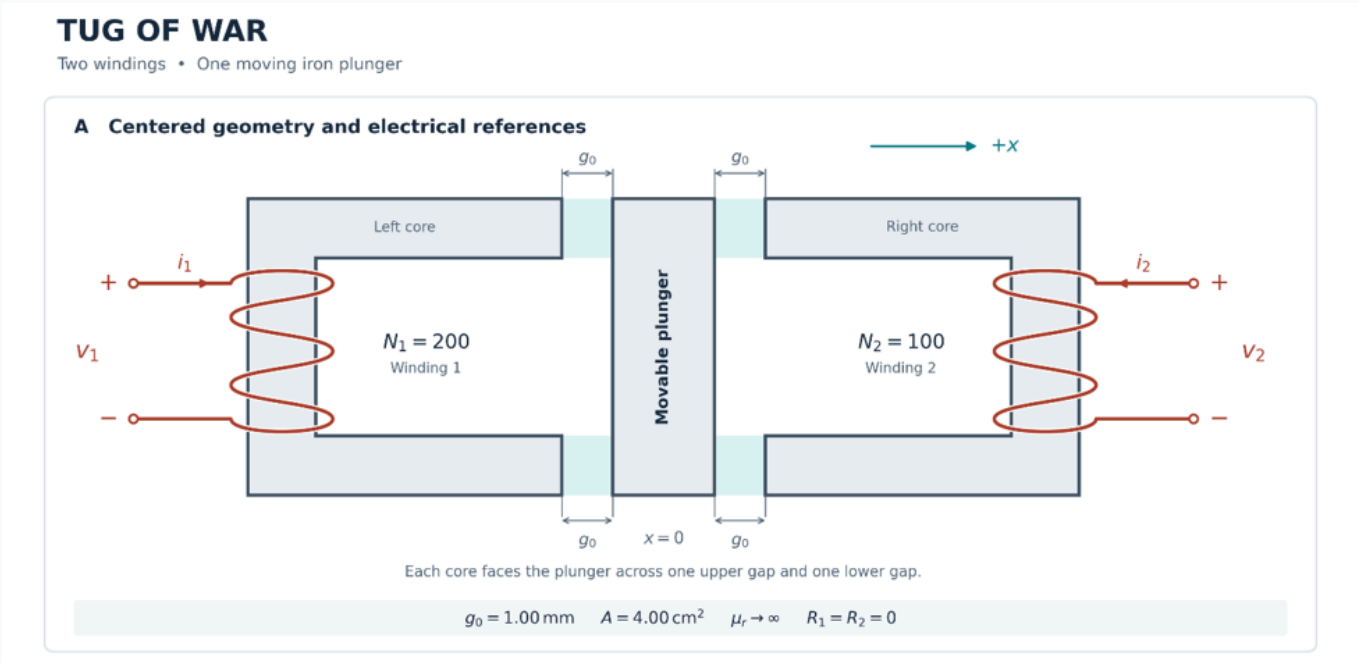
(c) Express $i_2(t)$ in terms of $V_2$, time, and the plunger position x(t).

(d) As the plunger moves left, explain how both the applied voltage and the changing inductance affect the current in winding 2.

(e) Must the plunger continue moving left? Can you determine whether it reverses before contacting a pole from the information given? Explain what additional information would be needed.

(f) Why does constant-voltage excitation imply neither constant current nor constant electromagnetic force?

# 8. Derive the electromagnetic force




**System description**

Two fixed C-shaped iron cores face a central movable iron plunger. Winding 1 is on the left core, and winding 2 is on the right core. Each core faces the plunger across two air gaps: one upper gap and one lower gap.

For horizontal motion, define the plunger displacement x as positive to the right. At x = 0, all four air gaps have length $g_0$.

The parameters are:

$N_1 = 200$ turns
$N_2 = 100$ turns
$g_0 = 1.00$ mm
$A = 4.00$ cm²
$\mu_0 = 4\pi \times 10^{-7}$ H/m

Use the following assumptions:

- All iron has infinite permeability.
- Both winding resistances are zero.
- Leakage, hysteresis, and eddy currents are negligible.
- Neglect fringing in the reluctance calculations.
- For horizontal motion, each gap has constant face area A.
- For vertical motion in Part 9, use the specified model for changing pole overlap.
- Neglect gravity, friction, and external mechanical forces unless a constraint is specified.
- Ideal electrical sources can supply or absorb energy.

Use the passive sign convention for each winding: positive current enters its positive voltage terminal.

Treat Parts 4, 6, 7, 9, 10, and 11 as separate experiments. Use the stated initial conditions for each experiment.

> **Referenced Part 4 conditions**
>
> The plunger is initially centered and at rest: x = 0 and $\frac{dx}{dt} = 0$. Ideal current sources maintain $i_1 = 1.00$ A and $i_2 = 1.90$ A.

Derive the net horizontal electromagnetic force:

$F_{em}(i_1, i_2, x)$.

Take rightward force as positive.

(a) Obtain an expression for the magnetic coenergy of the complete system.

(b) Differentiate the coenergy to obtain the force. State explicitly which quantities you hold constant during differentiation.

(c) Identify the separate leftward and rightward force contributions.

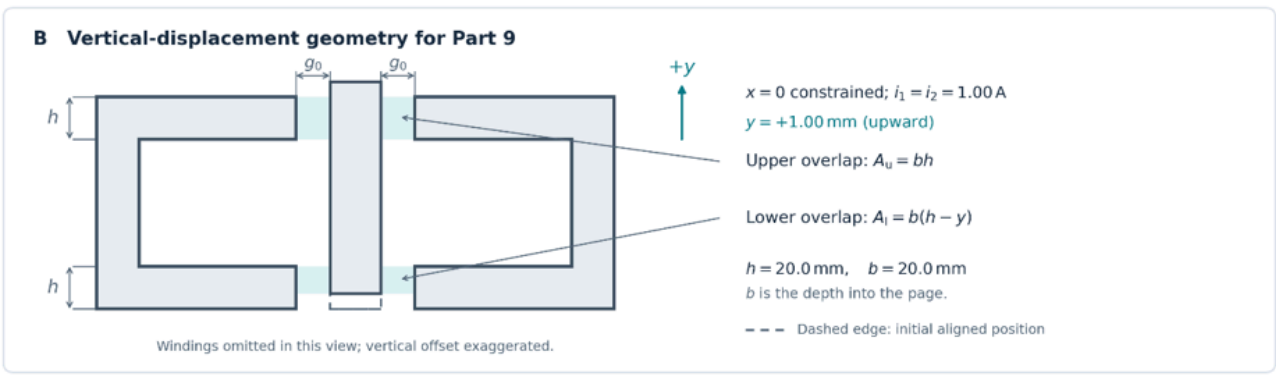
(d) Check your result independently using magnetic pressure at all four gap faces.

(e) Calculate the net force for the conditions in Part 4. Compare its direction with your earlier prediction.

(f) If either winding current is reversed, does that winding's core repel the plunger? Explain using both your force expression and physical reasoning.

(g) Does holding the currents constant during the coenergy differentiation mean that your force expression applies only to constant-current operation? Explain.

# 9. Perturb the plunger vertically




**System description**

Two fixed C-shaped iron cores face a central movable iron plunger. Winding 1 is on the left core, and winding 2 is on the right core. Each core faces the plunger across two air gaps: one upper gap and one lower gap.

For horizontal motion, define the plunger displacement x as positive to the right. At x = 0, all four air gaps have length $g_0$.

The parameters are:

$N_1 = 200$ turns
$N_2 = 100$ turns
$g_0 = 1.00$ mm
$A = 4.00$ cm²
$\mu_0 = 4\pi \times 10^{-7}$ H/m

Use the following assumptions:

- All iron has infinite permeability.
- Both winding resistances are zero.
- Leakage, hysteresis, and eddy currents are negligible.
- Neglect fringing in the reluctance calculations.
- For horizontal motion, each gap has constant face area A.
- For vertical motion in Part 9, use the specified model for changing pole overlap.
- Neglect gravity, friction, and external mechanical forces unless a constraint is specified.
- Ideal electrical sources can supply or absorb energy.

Use the passive sign convention for each winding: positive current enters its positive voltage terminal.

Treat Parts 4, 6, 7, 9, 10, and 11 as separate experiments. Use the stated initial conditions for each experiment.

Constrain horizontal motion at:

x = 0.

The guide supplies any required horizontal reaction but exerts no vertical force. Maintain equal constant winding currents:

$i_1 = i_2 = 1.00$ A.

Each pole face has vertical height h and depth b into the page:

h = 20.0 mm and b = 20.0 mm,

so its full area is bh = A.

Initially, the plunger's upper and lower ends are flush with the outer edges of the upper and lower poles. Define vertical displacement y as positive upward.

For an upward displacement satisfying $0 < y < h$, use the following overlap approximation on both sides:

$A_u = bh$

$A_l = b(h - y)$

Here $A_u$ is the upper overlap area and $A_l$ is the lower overlap area. All four horizontal gap lengths remain $g_0$.

The plunger is displaced upward by:

y = 1.00 mm

and held at rest.

(a) Update the magnetic equivalent circuit to account for the unequal upper and lower overlap areas.

(b) Derive the magnetic coenergy as a function of y.

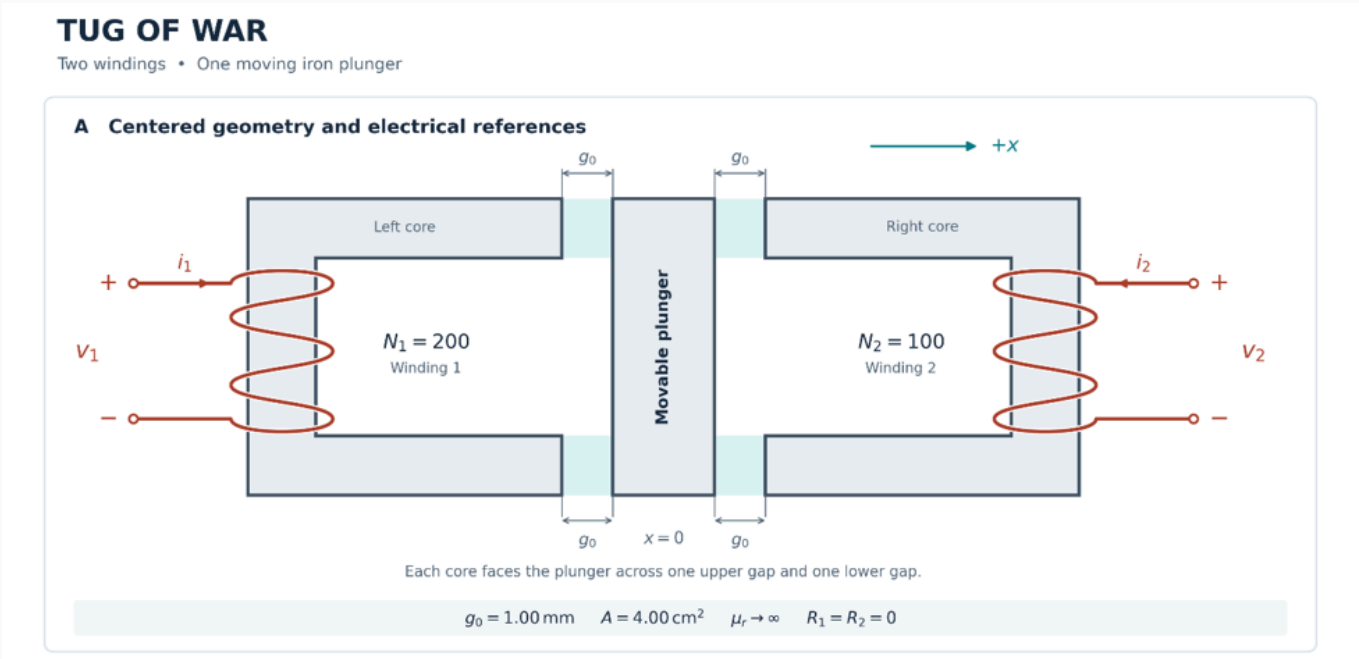
(c) Calculate the approximate vertical electromagnetic force. Predict the initial direction of motion when the vertical restraint is released.

(d) Would a downward displacement produce an upward or downward electromagnetic force? Is the magnetic force restoring or destabilizing with respect to vertical alignment?

(e) What vertical force would the earlier constant-area model predict? Explain why the two models give different predictions.

(f) Do equal winding currents imply equal horizontal attractions in this system? Explain the role of the horizontal guide.

# 10. Challenge extension — Change the electrical connections




**System description**

Two fixed C-shaped iron cores face a central movable iron plunger. Winding 1 is on the left core, and winding 2 is on the right core. Each core faces the plunger across two air gaps: one upper gap and one lower gap.

For horizontal motion, define the plunger displacement x as positive to the right. At x = 0, all four air gaps have length $g_0$.

The parameters are:

$N_1 = 200$ turns
$N_2 = 100$ turns
$g_0 = 1.00$ mm
$A = 4.00$ cm²
$\mu_0 = 4\pi \times 10^{-7}$ H/m

Use the following assumptions:

- All iron has infinite permeability.
- Both winding resistances are zero.
- Leakage, hysteresis, and eddy currents are negligible.
- Neglect fringing in the reluctance calculations.
- For horizontal motion, each gap has constant face area A.
- For vertical motion in Part 9, use the specified model for changing pole overlap.
- Neglect gravity, friction, and external mechanical forces unless a constraint is specified.
- Ideal electrical sources can supply or absorb energy.

Use the passive sign convention for each winding: positive current enters its positive voltage terminal.

Treat Parts 4, 6, 7, 9, 10, and 11 as separate experiments. Use the stated initial conditions for each experiment.

Consider a separate experiment. Initially, the plunger is centered and at rest, with:

$i_1 = 1.00$ A and $i_2 = 2.00$ A.

Compare two cases:

Case A: Ideal current sources maintain both currents at their initial values.

Case B: After establishing the initial currents at x = 0, disconnect the current sources and individually short-circuit each winding. Both winding resistances remain zero.

For each case, an external mechanism slowly displaces the plunger a small horizontal distance from the center. The electrical connections remain as specified while it is displaced. The plunger is then released from rest.

(a) Verify that the initial electromagnetic force at x = 0 is zero.

(b) Identify which electrical quantities remain constant in each case.

(c) Derive $i_1(x)$ and $i_2(x)$ for each case.

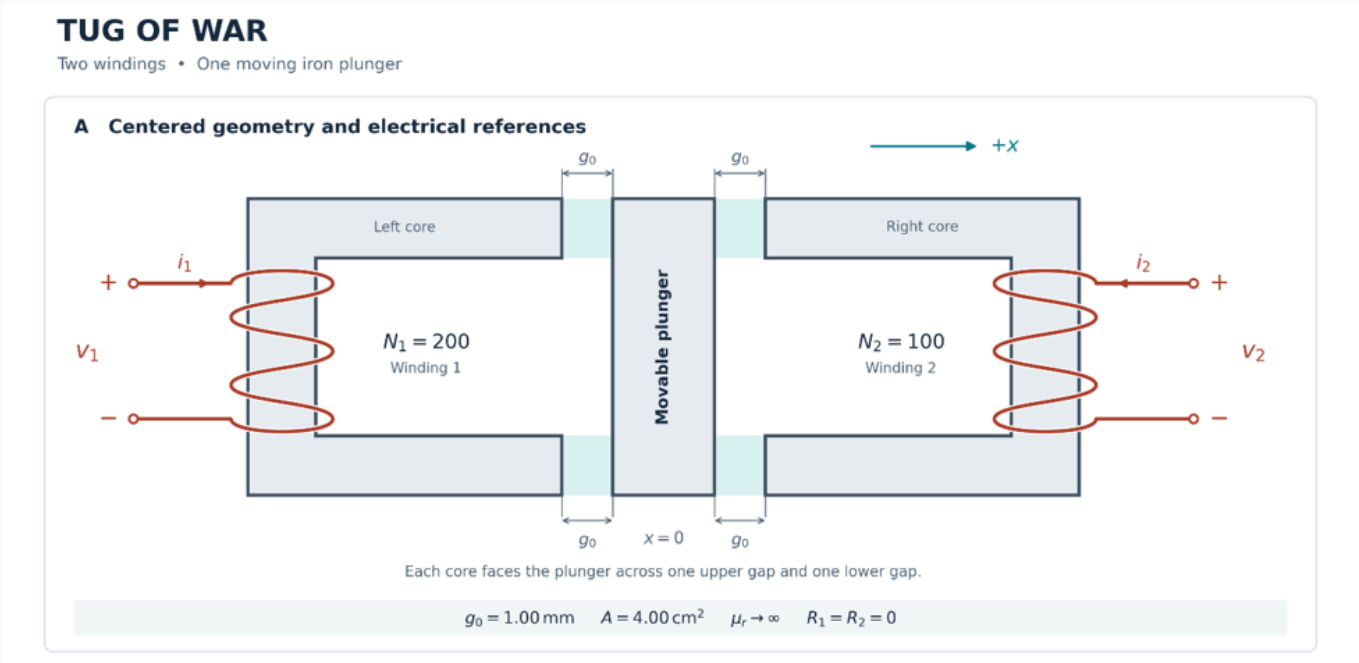
(d) Derive the net electromagnetic force as a function of position for each case.

(e) Classify the centered position as restoring, destabilizing, or neutral in each case.

(f) Explain how zero winding terminal voltage can coexist with changing current and a nonzero speed-voltage contribution.

(g) Explain why identical geometry and identical initial currents can lead to different responses after the electrical connections are changed.

# Final reflection




**System description**

Two fixed C-shaped iron cores face a central movable iron plunger. Winding 1 is on the left core, and winding 2 is on the right core. Each core faces the plunger across two air gaps: one upper gap and one lower gap.

For horizontal motion, define the plunger displacement x as positive to the right. At x = 0, all four air gaps have length $g_0$.

The parameters are:

$N_1 = 200$ turns
$N_2 = 100$ turns
$g_0 = 1.00$ mm
$A = 4.00$ cm²
$\mu_0 = 4\pi \times 10^{-7}$ H/m

Use the following assumptions:

- All iron has infinite permeability.
- Both winding resistances are zero.
- Leakage, hysteresis, and eddy currents are negligible.
- Neglect fringing in the reluctance calculations.
- For horizontal motion, each gap has constant face area A.
- For vertical motion in Part 9, use the specified model for changing pole overlap.
- Neglect gravity, friction, and external mechanical forces unless a constraint is specified.
- Ideal electrical sources can supply or absorb energy.

Use the passive sign convention for each winding: positive current enters its positive voltage terminal.

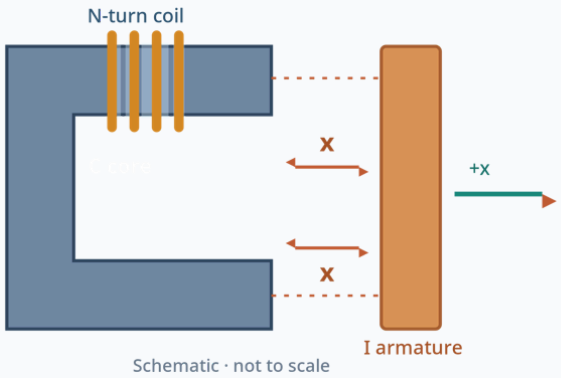
Treat Parts 4, 6, 7, 9, 10, and 11 as separate experiments. Use the stated initial conditions for each experiment.

Imagine receiving a different actuator diagram without these numbered questions.

What steps would you take to begin? Which physical laws and relationships would still apply, and which geometric or constitutive relationships would you need to derive again?

Identify one prediction that you revised while solving this problem. Explain what changed your reasoning.

# 1. Build the magnetic model.



Schematic · not to scale

**The C–I Core That Changes Jobs: One electromagnetic device, two opposite energy-conversion cycles**

An engineering team is developing a small reciprocating electromechanical device for a laboratory energy-conversion demonstrator. Its stationary member is a C-shaped magnetic core carrying one winding. A separate I-shaped iron armature slides toward and away from the C core. A crank attached to the armature connects the device to an external mechanical shaft.

The winding is connected to a bidirectional electronic converter and a DC bus. The converter can supply current to the winding or accept electrical energy returned by it. The engineering team wants to operate the very same hardware in two ways: first to deliver mechanical energy to the shaft, and then to absorb mechanical energy from an external prime mover and send energy back to the DC bus. Your task is to predict what happens at both ports of the device over one complete cycle.

*Figure 1. C–I magnetic structure. Each air gap has length x; positive x points away from the C core.*

**Test-bench specifications**

| | | | |
|---|---|---|---|
| Winding turns, N | 200 | Pole-face area, A | 400 mm² |
| Closed position | x = 2 mm | Open position | x = 6 mm |
| Commanded on-current | i = +2 A | Cycle duration | 40 ms |

Model the magnetic material as linear with negligible iron reluctance, leakage, and fringing. There are two identical air gaps in series; ignore winding resistance and mechanical losses. Use $\mu_0 = 4\pi \times 10^{-7}$ H/m. The converter enforces the current program, and the crank provides the prescribed position program. Treat changes of motion slope at the switching instants as idealized.

**Port conventions**

Take $i > 0$ entering the positive-voltage terminal, so $v = \frac{d\lambda}{dt}$ and $vi > 0$ represents electrical power entering the magnetic device. Take $x > 0$ away from the C core; $F_e$ is positive if the force of electric origin point toward increasing x. Thus, $F_e(\frac{dx}{dt}) > 0$ represents the mechanical power the electromagnetic field delivers to the shaft.

**Two experimental runs**

In Run M, the converter energizes the coil with the armature far from the core. The magnetic attraction then draws the armature inward against the shaft load. The winding is switched off before the armature is returned outward.

In Run G, an external prime mover first holds the armature close to the core while the winding is energized. The prime mover then pulls the armature outward against the magnetic attraction. The winding is switched off before the armature is returned inward. No field force is present during either unenergized return stroke.

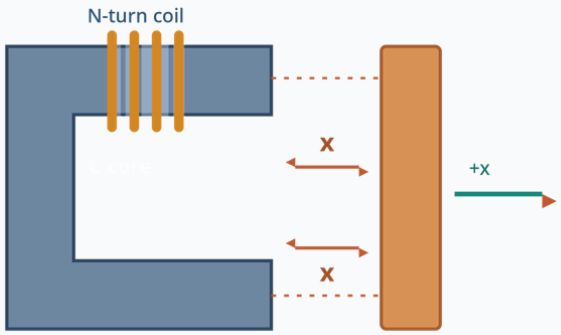
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| Leg | Time | Coil-current program | Run M: x | Run G: x |
|---|---|---|---|---|
| A → B | 0–4 ms | 0 → +2 A; x fixed | 6 mm | 2 mm |
| B → C | 4–20 ms | +2 A held constant | 6 → 2 mm | 2 → 6 mm |
| C → D | 20–24 ms | +2 → 0 A; x fixed | 2 mm | 6 mm |
| D → A | 24–40 ms | 0 A throughout | 2 → 6 mm | 6 → 2 mm |

### Your investigation — complete each task for BOTH runs

Beginning with the two air-gap reluctances, derive $R_m(x)$, $L(x)$, $\lambda(i,x)$, stored field energy $W_f(i,x)$, field coenergy $W'_f(i,x)$, and electromagnetic force $F_e(i,x)$. Give the force direction for $i \neq 0$, and explain whether reversing the coil current reverses this force.

# 2. Identify the four states.



Schematic · not to scale

**The C–I Core That Changes Jobs: One electromagnetic device, two opposite energy-conversion cycles**

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The winding is connected to a bidirectional electronic converter and a DC bus. The converter can supply current to the winding or accept electrical energy returned by it. The engineering team wants to operate the very same hardware in two ways: first to deliver mechanical energy to the shaft, and then to absorb mechanical energy from an external prime mover and send energy back to the DC bus. Your task is to predict what happens at both ports of the device over one complete cycle.

*Figure 1. C–I magnetic structure. Each air gap has length x; positive x points away from the C core.*

**Test-bench specifications**

| | | | |
|---|---|---|---|
| Winding turns, N | 200 | Pole-face area, A | 400 mm² |
| Closed position | x = 2 mm | Open position | x = 6 mm |
| Commanded on-current | i = +2 A | Cycle duration | 40 ms |

Model the magnetic material as linear with negligible iron reluctance, leakage, and fringing. There are two identical air gaps in series; ignore winding resistance and mechanical losses. Use $\mu_0 = 4\pi \times 10^{-7}$ H/m. The converter enforces the current program, and the crank provides the prescribed position program. Treat changes of motion slope at the switching instants as idealized.

**Port conventions**

Take i > 0 entering the positive-voltage terminal, so $v = \frac{d\lambda}{dt}$ and vi > 0 represents electrical power entering the magnetic device. Take x > 0 away from the C core; $F_e$ is positive if the force of electric origin point toward increasing x. Thus, $F_e(\frac{dx}{dt}) > 0$ represents the mechanical power the electromagnetic field delivers to the shaft.

**Two experimental runs**

In Run M, the converter energizes the coil with the armature far from the core. The magnetic attraction then draws the armature inward against the shaft load. The winding is switched off before the armature is returned outward.

In Run G, an external prime mover first holds the armature close to the core while the winding is energized. The prime mover then pulls the armature outward against the magnetic attraction. The winding is switched off before the armature is returned inward. No field force is present during either unenergized return stroke.

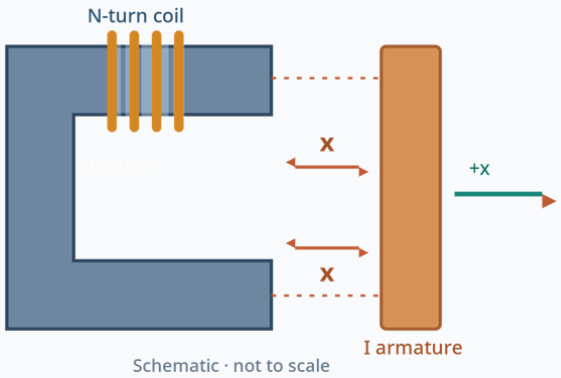
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| Leg | Time | Coil-current program | Run M: x | Run G: x |
|---|---|---|---|---|
| A → B | 0–4 ms | 0 → +2 A; x fixed | 6 mm | 2 mm |
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| C → D | 20–24 ms | +2 → 0 A; x fixed | 2 mm | 6 mm |
| D → A | 24–40 ms | 0 A throughout | 2 → 6 mm | 6 → 2 mm |

### Your investigation — complete each task for BOTH runs

At A, B, C, D, and A at 40 ms, determine x, i, λ, and Wf. Organize the results in a state table for each run. State which legs change stored field energy and which transfer mechanical energy.

# 3. Draw the three plane trajectories.



Schematic · not to scale

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*Figure 1. C–I magnetic structure. Each air gap has length x; positive x points away from the C core.*

**Test-bench specifications**

| | | | |
|---|---|---|---|
| Winding turns, N | 200 | Pole-face area, A | 400 mm² |
| Closed position | x = 2 mm | Open position | x = 6 mm |
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Model the magnetic material as linear with negligible iron reluctance, leakage, and fringing. There are two identical air gaps in series; ignore winding resistance and mechanical losses. Use $\mu_0 = 4\pi \times 10^{-7}$ H/m. The converter enforces the current program, and the crank provides the prescribed position program. Treat changes of motion slope at the switching instants as idealized.

**Port conventions**

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**Two experimental runs**

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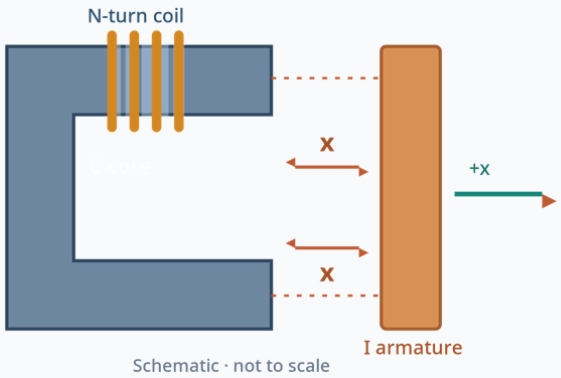
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### Your investigation — complete each task for BOTH runs

On the supplied axes, sketch the directed A → B → C → D → A paths in the λ–i, λ–x, and x–i planes. Use λ as the horizontal coordinate in the λ–i plane. Show both runs, identify A/B/C/D, and use the actual functional relationships rather than merely connecting endpoints with straight lines.

# 4. Reconstruct the waveforms.




Schematic · not to scale

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**Test-bench specifications**

| | | | |
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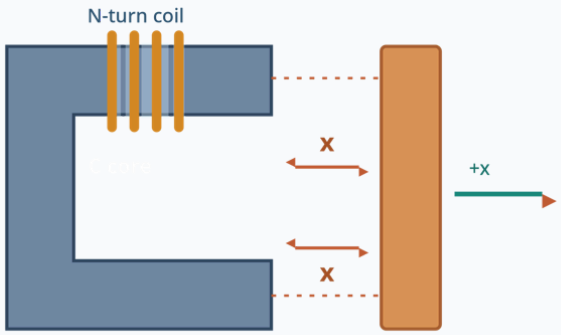
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### Your investigation — complete each task for BOTH runs

Plot x(t), λ(t), i(t), and terminal voltage v(t) over 0–40 ms. Derive the voltage for each leg separately, including the speed-voltage contribution during armature motion. Explain the polarity of v during B → C in both runs.

# 5. Account for the exchanged energy.



Schematic · not to scale

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**Test-bench specifications**

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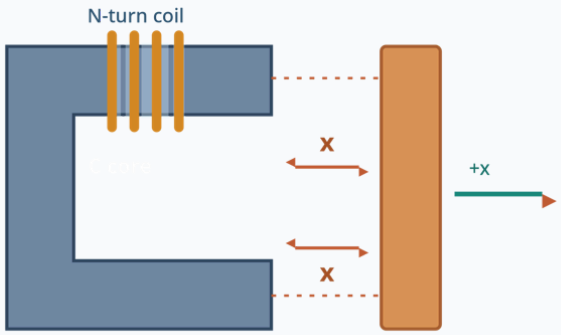
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### Your investigation — complete each task for BOTH runs

Define $E_FE(t) = \int_0^t v\, i\, dt$ as cumulative electrical energy entering the device and $E_FM(t) = \int_0^t F_e(\frac{dx}{dt})\, dt$ as cumulative mechanical energy leaving it. Calculate each quantity at A, B, C, D, and the final A; then overlay E_FE(t) and E_FM(t) on common axes for each run.

# 6. Close the energy balance and interpret.



Schematic · not to scale

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### Your investigation — complete each task for BOTH runs

Verify at B, C, D, and 40 ms that E_FE(t) − E_FM(t) = Wf(t) − Wf(0). Determine the net energy exchanged per cycle, identify which run is motor action and which is generator action, and relate the sign of net electrical energy to the direction and enclosed area of the λ–i loop. Explain why the area of the x–i loop is not mechanical energy.